

SU(3) Gauge Connection: Three-Qubit Encoding

Equivalence at the Level of Lie Algebra Representation and Covariant Derivative

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Abstract

We establish a precise equivalence between the standard UBT SU(3) gauge connection $A_\mu = A_\mu^a \lambda_a / 2$ and a three-qubit encoded gauge connection $A_\mu^{3q} = A_\mu^a G_a / 2$, where G_a are the Gell-Mann matrices lifted to the one-hot (color) subspace of $\mathcal{H}_{3q} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$. The equivalence is representation-level: both descriptions carry the same $\mathfrak{su}(3)$ Lie algebra and the same local gauge covariant derivative on the color subspace $\mathcal{H}_{\text{color}} = \text{span}\{|100\rangle, |010\rangle, |001\rangle\}$. The full three-qubit space contains an orthogonal leakage sector $\mathcal{H}_{\text{color}}^\perp$, which must be constrained or interpreted separately.

1 Setup and Notation

Status and scope

Scope: This file proves equivalence at the Lie algebra representation and covariant derivative level. It does *not* claim physical derivation of full QCD dynamics, confinement, anomaly cancellation, or the fermion representation content.

Canonical SU(3) source: Theorems G.A–G.D in canonical/interactions/sm_gauge.tex; the involution derivation in canonical/su3_derivation/su3_from_involutions.tex.

Three-qubit source: Geometric analysis in research_tracks/triqubit_su3_geometry.md; one-hot embedding in research_tracks/THEORY_COMPARISONS/su3_qubit_mapping/su3_qubit_core/mapping.py.

Conventions. We follow the standard normalisation $\text{Tr}[\lambda_a \lambda_b] = 2\delta_{ab}$ for Gell-Mann matrices. Indices $a, b, c \in \{1, \dots, 8\}$ label adjoint generators; $i, j \in \{1, 2, 3\}$ label the fundamental color triplet. The strong coupling is g (or g_s when disambiguation is needed).

2 Definitions

Definition 2.1 (Three-qubit Hilbert space). The *three-qubit Hilbert space* is

$$\mathcal{H}_{3q} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^8,$$

with computational basis $\{|q_1 q_2 q_3\rangle \mid q_i \in \{0, 1\}\}$.

Definition 2.2 (Color subspace). The *color subspace* (one-hot sector) is

$$\mathcal{H}_{\text{color}} = \text{span}\{|100\rangle, |010\rangle, |001\rangle\} \subset \mathcal{H}_{3q}.$$

The three basis states are identified with the three quark color charges:

$$\begin{aligned} |100\rangle &\longleftrightarrow |r\rangle \quad (\text{red}), \\ |010\rangle &\longleftrightarrow |g\rangle \quad (\text{green}), \\ |001\rangle &\longleftrightarrow |b\rangle \quad (\text{blue}). \end{aligned}$$

Definition 2.3 (Color projector). The orthogonal projector onto $\mathcal{H}_{\text{color}}$ is

$$P_{\text{color}} = |100\rangle\langle 100| + |010\rangle\langle 010| + |001\rangle\langle 001| \in \text{End}(\mathbb{C}^8).$$

Definition 2.4 (Encoded generators). For each Gell-Mann matrix λ_a ($a = 1, \dots, 8$) acting on $\mathbb{C}^3$, the *encoded generator* $G_a \in \text{End}(\mathbb{C}^8)$ is defined by

$$G_a = P \lambda_a P^\dagger,$$

where $P : \mathbb{C}^3 \rightarrow \mathbb{C}^8$ is the isometric one-hot embedding

$$P = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

mapping $|r\rangle \mapsto |100\rangle$, $|g\rangle \mapsto |010\rangle$, $|b\rangle \mapsto |001\rangle$.

Equivalently, G_a acts as λ_a on $\mathcal{H}_{\text{color}}$ and as zero on $\mathcal{H}_{\text{color}}^\perp$:

$$G_a|_{\mathcal{H}_{\text{color}}} = \lambda_a, \quad G_a|_{\mathcal{H}_{\text{color}}^\perp} = 0.$$

Remark 2.5 (Generator normalisation). The matrices G_a lift the full Gell-Mann matrices, not the conventionally normalised Lie generators $\lambda_a/2$. Consequently $[G_a, G_b] = 2if_{abc}G_c$, while gauge connections and exponentials use $G_a/2$. This distinction is kept explicit below.

Definition 2.6 (Encoded gauge connection and covariant derivative). Given real gauge functions $A_\mu^a(x)$, define

$$A_\mu^{3q}(x) = \sum_{a=1}^8 A_\mu^a(x) \frac{G_a}{2} \in \text{End}(\mathbb{C}^8),$$

and the *encoded covariant derivative*

$$D_\mu^{3q} = \partial_\mu + ig A_\mu^{3q}.$$

QCD color	Standard $\mathbb{C}^3$ basis	One-hot qubit state	Qubit basis index
Red	$e_1 = (1, 0, 0)^T$	$ 100\rangle$	4
Green	$e_2 = (0, 1, 0)^T$	$ 010\rangle$	2
Blue	$e_3 = (0, 0, 1)^T$	$ 001\rangle$	1

Table 1: Basis mapping between standard QCD color space and one-hot three-qubit states. Qubit basis index uses binary encoding $4q_1 + 2q_2 + q_3$ (0-indexed).

Index a	Standard generator $\frac{\lambda_a}{2}$	Encoded generator $G_a = P\lambda_a P^\dagger$
1	$r \leftrightarrow g$ off-diagonal, real	$P_{\text{color}}(\lambda_1)P_{\text{color}}$, acts on $\{ 100\rangle, 010\rangle\}$
2	$r \leftrightarrow g$ off-diagonal, imaginary	idem, imaginary
3	r - g color isospin diagonal	$P_{\text{color}} \text{diag}(1, -1, 0) P_{\text{color}}$
4	$r \leftrightarrow b$ off-diagonal, real	$P_{\text{color}}(\lambda_4)P_{\text{color}}$, acts on $\{ 100\rangle, 001\rangle\}$
5	$r \leftrightarrow b$ off-diagonal, imaginary	idem, imaginary
6	$g \leftrightarrow b$ off-diagonal, real	$P_{\text{color}}(\lambda_6)P_{\text{color}}$, acts on $\{ 010\rangle, 001\rangle\}$
7	$g \leftrightarrow b$ off-diagonal, imaginary	idem, imaginary
8	color hypercharge diagonal	$P_{\text{color}} \text{diag}(1, 1, -2)/\sqrt{3} P_{\text{color}}$

Table 2: Generator mapping. $P_{\text{color}}(\lambda_a)P_{\text{color}}$ denotes the operator $P\lambda_a P^\dagger$ on $\mathbb{C}^8$, which acts as λ_a on $\mathcal{H}_{\text{color}}$ and vanishes on $\mathcal{H}_{\text{color}}^\perp$.

3 Basis and Generator Mapping

3.1 Color Basis Mapping

3.2 Generator Mapping

4 Proof of Commutation Relations

Theorem 4.1 (Encoded generators satisfy $\mathfrak{su}(3)$ on $\mathcal{H}_{\text{color}}$). *For all $a, b \in \{1, \dots, 8\}$,*

$$[G_a, G_b] = 2if_{abc} G_c \quad \text{on } \mathcal{H}_{\text{color}},$$

where f_{abc} are the totally antisymmetric $\mathfrak{su}(3)$ structure constants.

Proof. Let $|v\rangle \in \mathcal{H}_{\text{color}}$ be arbitrary. Write $|v\rangle = P\mathbf{v}$ for a unique $\mathbf{v} \in \mathbb{C}^3$. Since $P^\dagger P = I_3$ (isometry),

$$\begin{aligned} G_a G_b |v\rangle &= (P\lambda_a P^\dagger)(P\lambda_b P^\dagger)P\mathbf{v} \\ &= P\lambda_a (P^\dagger P)\lambda_b \mathbf{v} \\ &= P\lambda_a \lambda_b \mathbf{v}. \end{aligned}$$

Therefore

$$[G_a, G_b]|v\rangle = P[\lambda_a, \lambda_b]\mathbf{v} = P(2if_{abc}\lambda_c)\mathbf{v} = 2if_{abc} G_c |v\rangle,$$

using the standard $\mathfrak{su}(3)$ relation $[\lambda_a, \lambda_b] = 2if_{abc}\lambda_c$ in the last step.

For $|v\rangle \in \mathcal{H}_{\text{color}}^\perp$: $G_a |v\rangle = P\lambda_a P^\dagger |v\rangle = 0$ (since $P^\dagger |v\rangle = 0$), so $[G_a, G_b]|v\rangle = 0 = 2if_{abc} G_c |v\rangle$. The relation holds on all of $\mathcal{H}_{3q}$. $\square$

Remark 4.2. The commutation relations hold on the entire $\mathbb{C}^8$ (both $\mathcal{H}_{\text{color}}$ and its complement), but the action is trivial on $\mathcal{H}_{\text{color}}^\perp$. The algebra is realised faithfully on $\mathcal{H}_{\text{color}} \cong \mathbb{C}^3$.

5 Gauge Transformation Equivalence

Theorem 5.1 (Gauge transformations are compatible). *Let $\alpha^a(x)$ be real gauge parameters. Define*

$$U(x) = \exp(i\alpha^a(x)\frac{\lambda_a}{2}) \in SU(3), \quad U_{3q}(x) = \exp(i\alpha^a(x)\frac{G_a}{2}) \in U(8).$$

Then:

1. $U_{3q}(x)|_{\mathcal{H}_{\text{color}}}$ is unitarily equivalent to $U(x)$ via P :

$$U_{3q}(x)P = PU(x).$$

2. $U_{3q}(x)|_{\mathcal{H}_{\text{color}}^\perp} = \text{Id}|_{\mathcal{H}_{\text{color}}^\perp}$.

Proof. For part (1): Since $G_a P = P\lambda_a$ (from Definition 2.4) and λ_a matrices commute with this embedding, the exponential satisfies

$$\exp(i\alpha^a \frac{G_a}{2})P = P \exp(i\alpha^a \frac{\lambda_a}{2}),$$

which follows by expanding the matrix exponential and using $G_a P = P\lambda_a$ inductively (each power $(G_a)^n P = P(\lambda_a)^n$, verified by applying $P^\dagger P = I_3$ at each step).

For part (2): $G_a|_{\mathcal{H}_{\text{color}}^\perp} = 0$, so $\exp(i\alpha^a G_a/2)|_{\mathcal{H}_{\text{color}}^\perp} = e^0 = \text{Id}$. $\square$

6 Covariant Derivative Equivalence

Theorem 6.1 (Covariant derivatives are equivalent on $\mathcal{H}_{\text{color}}$). *Let $\psi(x) \in \mathcal{H}_{\text{color}}$ be a color state (a section of the color bundle). Write $\psi = P\varphi$ with $\varphi \in \mathbb{C}^3$. Then*

$$P^\dagger(D_\mu^{3q}\psi) = D_\mu\varphi,$$

where $D_\mu = \partial_\mu + igA_\mu^a \frac{\lambda_a}{2}$ is the standard QCD covariant derivative on $\mathbb{C}^3$.

Proof.

$$\begin{aligned} D_\mu^{3q}\psi &= \partial_\mu(P\varphi) + igA_\mu^a \frac{G_a}{2}(P\varphi) \\ &= P(\partial_\mu\varphi) + igA_\mu^a P \frac{\lambda_a}{2}\varphi \\ &= P\left(\partial_\mu\varphi + igA_\mu^a \frac{\lambda_a}{2}\varphi\right) = P(D_\mu\varphi). \end{aligned}$$

Applying $P^\dagger$ to both sides and using $P^\dagger P = I_3$ gives the claim. $\square$

Corollary 6.2 (Gauge transformation of D_μ^{3q}). *Under a local gauge transformation $U_{3q}(x)$, the encoded covariant derivative transforms as*

$$D_\mu^{3q} \longrightarrow U_{3q} D_\mu^{3q} U_{3q}^\dagger \quad \text{on } \mathcal{H}_{\text{color}},$$

consistent with the standard transformation law $D_\mu \rightarrow UD_\mu U^\dagger$ via the equivalence of Theorem 6.1.

7 Main Equivalence Theorem

Main Theorem: SU(3) Gauge–Qubit Equivalence

Theorem 7.1 (SU(3) gauge–three-qubit equivalence). *Let $\mathcal{H}_{\text{color}} = \text{span}\{|100\rangle, |010\rangle, |001\rangle\} \subset \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$. Then:*

1. **Generator equivalence.** *The map $\lambda_a \mapsto G_a = P\lambda_a P^\dagger$ is a faithful representation of $\mathfrak{su}(3)$ on $\mathcal{H}_{\text{color}}$, i.e. $[G_a, G_b] = 2if_{abc}G_c$ on $\mathcal{H}_{\text{color}}$.*
2. **Gauge field equivalence.** *The encoded gauge connection $A_\mu^{3q} = A_\mu^a G_a/2$ restricted to $\mathcal{H}_{\text{color}}$ is isomorphic to the standard connection $A_\mu = A_\mu^a \lambda_a/2$ via the isometry P .*
3. **Covariant derivative equivalence.** *The encoded derivative D_μ^{3q} restricted to $\mathcal{H}_{\text{color}}$ is unitarily equivalent to the standard QCD covariant derivative D_μ :*

$$P^\dagger D_\mu^{3q} P = D_\mu.$$

4. **Gauge transformation equivalence.** *Local SU(3) gauge transformations act consistently on both descriptions: $U_{3q}(x)P = P U(x)$.*

The equivalence is representation-level: it identifies the two descriptions as the same $\mathfrak{su}(3)$ gauge structure realized in different but isometric carrier spaces.

8 Leakage Sector Analysis

8.1 Structure of the Leakage Sector

The orthogonal complement of $\mathcal{H}_{\text{color}}$ in $\mathcal{H}_{3q}$ is the five-dimensional space

$$\mathcal{H}_{\text{color}}^\perp = \text{span}\{|000\rangle, |011\rangle, |101\rangle, |110\rangle, |111\rangle\}.$$

Its physical content is:

- $|000\rangle$: vacuum (no color excitation);
- $|011\rangle, |101\rangle, |110\rangle$: two-color (di-quark) excitations;
- $|111\rangle$: three-color (fully saturated) excitation.

8.2 Gauge Action on the Leakage Sector

All encoded generators G_a vanish on $\mathcal{H}_{\text{color}}^\perp$ (Definition 2.4), so gauge transformations act as the identity there. The leakage sector is gauge-invariant but physically distinct from the color subspace.

8.3 Interpretation and Constraint Options

Three interpretations are available; they should not be conflated:

1. **Hard projection (physical constraint):** Impose $P_{\text{color}}|\psi\rangle = |\psi\rangle$ as a superselection rule. This recovers standard QCD color quantum numbers; the leakage sector is unphysical.
2. **Auxiliary UBT degrees of freedom:** In the UBT context, the five leakage states may be reinterpreted as auxiliary biquaternionic degrees of freedom — e.g. the vacuum and multi-excitation modes of the Θ field beyond the fundamental color triplet. This interpretation is **speculative** and requires additional derivation from the UBT action; it is not established here.
3. **Quantum simulation context:** When using the three-qubit space as a quantum-simulation register for $\text{SU}(3)$ dynamics, the leakage sector must be actively suppressed (e.g. by energy penalty terms or by an external error-correction layer) to maintain faithful simulation. The one-hot encoding itself detects single- X_i/Y_i leakage but is not a quantum error-correcting code; see `canonical/interactions/gap_su3_triqubit_qec.tex`.

Warning: limits of the equivalence

The equivalence of Theorem 7.1 holds **strictly on** $\mathcal{H}_{\text{color}}$.

Do **not** claim:

- that the full Hilbert space $\mathcal{H}_{3q} = \mathbb{C}^8$ is identical to $\text{SU}(3)$ or to any physical color Hilbert space;
- that the three-qubit construction provides a physical derivation of QCD until dynamics (Yang-Mills action), confinement, anomaly cancellation, and fermion representations are addressed;
- that the leakage sector $\mathcal{H}_{\text{color}}^\perp$ has a canonical UBT interpretation without further derivation;
- that the one-hot sector is a quantum error-correcting code, or that its code-subspace form is evidence that physical reality is a simulation.

9 Relation to Canonical UBT $\text{SU}(3)$ Derivation

The canonical UBT derivation of $\text{SU}(3)_c$ proceeds via the involution structure of $\mathbb{C} \otimes \mathbb{H}$:

$$V_c = \ker(P_2 + \mathbf{1}) \cong \mathbb{C}^3, \quad \text{SU}(3) = \text{Aut}(V_c, \langle \cdot, \cdot \rangle),$$

(Theorems G.A–G.D; `canonical/su3_derivation/`). The present encoding identifies $\mathcal{H}_{\text{color}} \cong \mathbb{C}^3$ via a different route (one-hot embedding in $\mathcal{H}_{3q}$), but arrives at the same $\mathfrak{su}(3)$ algebra acting on the same three-dimensional complex vector space.

The two constructions are therefore **equivalent representations of the same Lie algebra**, realized via isometric embeddings of $\mathbb{C}^3$ into (different) ambient eight-dimensional spaces:

$$\begin{aligned} V_c &= \ker(P_2 + \mathbf{1}) \subset \mathbb{C} \otimes \mathbb{H} \cong \mathbb{C}^8, \\ \mathcal{H}_{\text{color}} &= \text{range}(P) \subset \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^8. \end{aligned}$$

Both identify the same group $SU(3)$; neither replaces the canonical derivation. For the algebraic bridge between the two ambient $\mathbb{C}^8$ spaces see the bridge document `canonical/bridges/su3_gau`